

LE NHAT TRUONG

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[Intruong.is-a.dev](https://github.com/letruong-is-a-dev) | [GitHub: lenhattruong-blip](https://github.com/lehattruong-blip)

TECHNICAL PROFILE

Final-year Integrated Circuit Engineering student seeking an Embedded Systems internship. Hands-on experience in STM32 PCB design and bring-up, ESP32-S3 sensor integration, and Python-based measurement analysis, with additional experience in FPGA implementation and waveform debugging.

EDUCATION

Ho Chi Minh City University of Foreign Languages and Information Technology (HUFLIT)

Major in Integrated Circuit Engineering (Ky thuật vi mạch) | **GPA: 3.63/4.00**

2023 - Present

Relevant Coursework: Embedded Systems, Embedded Programming, Microprocessors and Microcontrollers, Digital

Logic Design, Internet of Things, Intelligent Systems

Additional Training: Physical Design - Introductory Online Mini-course

SHTP Training Center & Tresemi | Certificate of Completion | Dec 2025

TECHNICAL SKILLS

Embedded and Hardware: STM32L4, ESP32-S3, schematic/PCB design, board bring-up, SPI, I2C, UART, SWD, FreeRTOS, RS485/Modbus RTU, MQTT/JSON

Programming and HDL: C, Python, Verilog

Data Analysis: Python, pandas, NumPy; measurement-data processing and visualization

Engineering Tools: Altium Designer, STM32CubeIDE, Arduino IDE, Vivado, Proteus; functional simulation and waveform analysis

TECHNICAL EXPERIENCE AND PROJECTS

STM32L4 Low-Power Hardware and Measurement Analysis

Feb - Aug 2026

Altium Designer | STM32L4P5 | INA228 | Statistical Analysis | [Project](#)

- Designed the STM32L4P5 extension for a team graduation thesis: a separate 2-layer power board and 4-layer MCU board, from schematic to PCB layout in Altium.
- Integrated SPI, I2C, UART and SWD interfaces; documented voltage/current measurements on the fabricated STM32L4P5 hardware.
- Completed physical bring-up successfully on the first fabricated revision.
- Separately analyzed the team's STM32L432 reference-board dataset using permutation tests and Benjamini-Hochberg correction to assess Buck-LDO power differences across voltage and operating conditions.

ESP32-S3 Greenhouse Sensor Integration

Dec 2025 - Mar 2026

ESP32-S3 | Arduino IDE | FreeRTOS | RS485/Modbus RTU | MQTT/JSON | [Project](#)

- Migrated sensor acquisition from a Raspberry Pi 5-based design to ESP32-S3, aiming to simplify the system and reduce cost and power demand; teammates handled the app and automation logic.
- Integrated industrial soil, environmental and CO2 sensors through RS485 Modbus RTU, I2C and UART using existing device libraries.
- Used three sensor tasks and one MQTT task with a shared-data mutex, JSON telemetry and MQTT retry handling; integrated the subsystem into a working physical setup.

RV32IM CPU - FPGA Implementation and Verification

Feb 2025 - Present

Vivado | XSim | Timing and Resource Analysis | Waveform Debugging | [Project](#)

- Ran FPGA implementation for a team-developed RV32IM CPU targeting xc7z030ffg676-1 and reviewed timing/resource reports across nine adder-multiplier configurations.
- Analyzed simulation waveforms to investigate control-state sequencing, program-counter progression and ALU behavior.
- Prepared selected implementation evidence for a source-free portfolio; physical FPGA-board operation has not been demonstrated.

Vietnamese Traffic-Sign Detection with YOLO26s

May - Aug 2026

Python | OpenCV | YOLO26s | Dataset Preparation and Inference | [Project](#)

- Led dataset sourcing, analysis and consolidation; the team report records 10,172 images, 17,860 annotated objects and 66 configured classes.
- Co-developed Python recognition and tested inference with teammate Thanh, using weights trained by another member. Team-reported test mAP@0.5: 0.987; two classes had no evaluation samples.